The Separable C^*-Algebra Case of BCP

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Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving that a commutative unital C^*-algebra $C(K)$ has BCP/UBCP exactly when K has a countable π-basis, the authors ask for a noncommutative version:

> $\{U_\alpha\}_{\alpha\in A}$ of K is called a π-basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].
>
> In the setting of commutative C^*-algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.
>
> **Corollary 3.9.** *Let $\mathcal{A}$ be a unital commutative C^*-algebra. Then the followings are equivalent:*
>
> (1) *$\mathcal{A}$ has the BCP;*
> (2) *the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π-basis;*
> (3) *$\mathcal{A}$ has the UBCP.*
>
> In particular, there exist commutative C^*-algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].
>
> We conclude this section by the following question.
>
> **Question 3.** *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^*-algebras $\mathcal{A}$ having BCP/UBCP.*

Statement

Theorem 1. *Every separable normed space has UBCP. Consequently every separable C^*-algebra has UBCP.*

Corollary 1. *If A is a separable C^*-algebra, then the following hold:*

$$A \text{ has UBCP}, \qquad A \text{ has BCP}, \qquad \mathrm{Prim}(A) \text{ has a countable } \pi\text{-base}.$$

In particular, the countably generated C^*-algebra case of the 2311.14257 question is settled positively.

Idea of the Proof

The proof separates into two familiar facts. First, separable normed spaces have a countable dense set in the unit sphere. If d is close to a unit vector x, then x lies in the ball centered at $2d$ of radius

3/2; this ball avoids the origin because its center has norm 2. This is a uniform ball cover of the unit sphere.

Second, if A is a separable C^*-algebra, then $\mathrm{Prim}(A)$ is second countable. A countable dense set in A, together with rational thresholds, gives a countable base by primitive near-norm sets

$$\{P \in \mathrm{Prim}(A) : \|a + P\| > t\}.$$

Thus the primitive pi-base condition from the previous obstruction packet is automatic in the separable case.

Proof

Lemma 1. *Every separable normed space X has UBCP.*

Proof. If $X = \{0\}$, the unit sphere is empty and there is nothing to prove. Assume $X \neq \{0\}$. Let $D \subseteq S_X$ be countable and 1/4-dense in S_X. For each $d \in D$, put

$$c_d = 2d.$$

Then $\|c_d\| = 2$, so the open ball $B(c_d, 3/2)$ avoids the origin. If $x \in S_X$, choose $d \in D$ with $\|x - d\| < 1/4$. Then

$$\|x - c_d\| = \|x - 2d\| \leq \|x - d\| + \|d\| < \frac{1}{4} + 1 < \frac{3}{2}.$$

Hence

$$S_X \subseteq \bigcup_{d \in D} B(2d, 3/2),$$

a countable equal-radius cover by balls off the origin. □

Proof of Theorem 1. Apply Lemma 1 to the underlying normed space of the separable C^*-algebra. □

Lemma 2. *If A is a separable C^*-algebra, then* $\mathrm{Prim}(A)$ *is second countable. In particular it has a countable π-base.*

Proof. Let D be a countable dense subset of A, and let

$$\mathcal{B} = \big\{U(d, q) : d \in D,\ q \in \mathbb{Q},\ q > 0\big\},$$

where

$$U(d, q) = \{P \in \mathrm{Prim}(A) : \|d + P\| > q\}.$$

Each $U(d, q)$ is open: as in the primitive pi-base obstruction packet,

$$U(d, q) = \{P : (d^*d - q^2)_+ \notin P\}.$$

We show that $\mathcal{B}$ is a base. Let $U_I = \{P : I \not\subseteq P\}$ be a basic open set corresponding to a nonzero closed two-sided ideal I, and let $P_0 \in U_I$. Choose $b \in I$ with $b \notin P_0$. Replacing b by b^*b and rescaling if necessary, assume $b \geq 0$ and $\|b + P_0\| > 0$. Choose $0 < \eta < \|b + P_0\|/3$, and choose $d \in D$ with

$$\|d - b\| < \eta.$$

Pick a rational q with

$$\eta < q < \|b + P_0\| - \eta.$$

Then $P_0 \in U(d, q)$, because

$$\|d + P_0\| \geq \|b + P_0\| - \eta > q.$$

If $P \notin U_I$, then $I \subseteq P$, so $b \in P$, and hence

$$\|d + P\| \leq \|d - b\| < \eta < q.$$

Thus $U(d, q) \subseteq U_I$. Therefore $\mathcal{B}$ refines every basic open neighborhood, so it is a countable base for $\mathrm{Prim}(A)$. □

Proof of Corollary 1. The UBCP assertion is Theorem 1, and BCP follows immediately from UBCP. Lemma 2 gives a countable π-base for $\mathrm{Prim}(A)$.

Finally, a countably generated C^*-algebra is separable: the $*$-polynomials with coefficients in $\mathbb{Q} + i\mathbb{Q}$ in a countable generating set form a countable dense subset of the generated algebra. Hence the countably generated case follows from the separable case. □

Relation to the General Problem

This packet settles the separable/countably generated subcase, but it also explains why the remaining problem is genuinely nonseparable. The previous primitive pi-base obstruction says that BCP forces $\mathrm{Prim}(A)$ to have a countable π-base for every C^*-algebra. In separable algebras this condition is automatic, and the Banach-space separability argument supplies UBCP.

For nonseparable algebras, neither side is automatic. Commutative examples show that nonseparable algebras can still have BCP when the underlying topological space has a countable π-base, while the primitive pi-base obstruction rules out many large ideal lattices. Thus the remaining frontier is the nonseparable, countable-π-base case.

Verification and Search Bounds

The run indexes were searched for arXiv:2311.14257 together with separable C*-algebra, countably generated C*-algebra, UBCP, BCP, $\mathrm{Prim}(A)$, second countable primitive ideal space, and countable π-base. Existing packets contained the primitive pi-base necessary condition and several continuous field/corona partial results, but no packet recorded the separable/countably generated classification.

Human Review Focus

This is a low-risk packet. Check the constants in Lemma 1 and the refinement argument in Lemma 2.

References

[1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.

[2] Jacques Dixmier, *C^*-algebras*, North-Holland, 1977.